

Genome-wide fusion-hotspot analysis of msk_impact_50k_2026

Abstract

This manuscript synthesizes a genome-wide fusion-hotspot cohort scan of msk_impact_50k_2026: 3919 genes carried at least one structural-variant record in the cohort, of which 544 passed the ≥ 5 -distinct-patient recurrence gate. All 544 gated genes were attempted: 10 using hand-curated gene configs and 514 auto-configured, with 20 gated-in gene(s) left unresolvable; 524 of the 544 attempted genes were successfully analyzed with the full registered algorithm suite. 4 genes reached genome-wide Benjamini-Hochberg FDR significance ($q < 0.05$): NTRK3 (58 events, 98.2% in-frame, 96.5% domain-retained, Fisher $p=0.0695489$, $q=9.00216e-09$); ROS1 (122 events, 72.7% in-frame, 90.7% domain-retained, Fisher $p=0.214047$, $q=0.00024396$); ETV6 (90 events, 81.0% in-frame, 77.3% domain-retained, Fisher $p=5.12326e-06$, $q=0.00257187$); FLI1 (118 events, 68.2% in-frame, 94.0% domain-retained, Fisher $p=0.0687101$, $q=0.00904727$). 15 additional genes form a highly ranked non-FDR-significant tier flagged for targeted follow-up (see Honorable mentions, below).

Methods

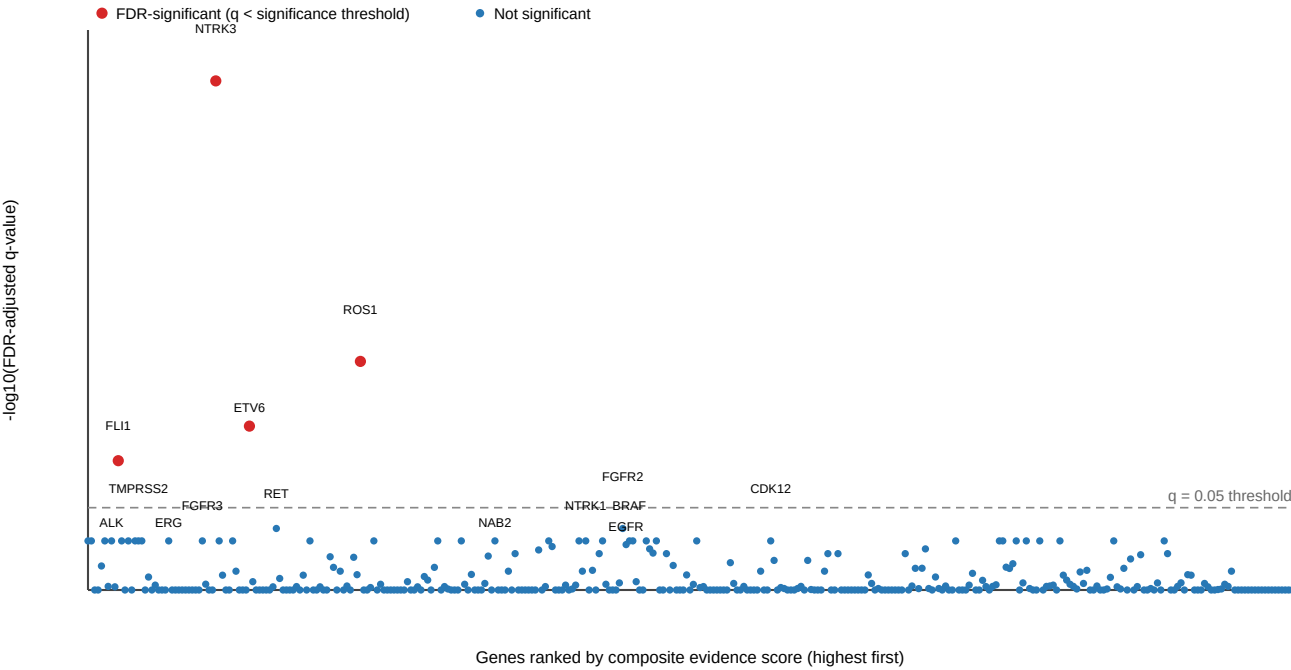
Structural-variant records were retrieved from the msk_impact_50k_2026 cBioPortal study and gated to genes with at least 5 distinct patient(s) carrying a structural-variant record (544 of 3919 genes passed the gate). All 544 gated genes were attempted: 10 using hand-curated gene configs and 514 auto-configured from Genome Nexus canonical-transcript/Pfam-domain data, with 20 gated-in gene(s) left unresolvable; 524 of the 544 attempted genes were successfully analyzed. Each successfully analyzed gene was run through the algorithm suite recorded for this scan (composite_score, confidence_stats, cutpoint_detection, domain_disruption, domain_retention, exon_retention, expression_association, frequency, genomic_position_recurrence, joint_partner, mechanistic_interpretation, mutation_cooccurrence, window_detection). Domain-retention and domain-disruption significance were assessed per gene with Fisher's exact test and a breakpoint-position permutation test; the resulting p-values across the 359 genes that produced at least one computable p-value were jointly corrected with Benjamini-Hochberg false-discovery-rate correction at $q < 0.05$.

Results

Genome-wide summary

Genome-wide summary of 359 scanned genes with an FDR-adjusted q-value, ranked left-to-right by composite evidence score; the dashed line marks the $q=0.05$ significance threshold, with 4 genes above it.

Genome-wide fusion-hotspot summary: FDR significance vs. composite evidence rank



FDR-significant and honorable-mention genes

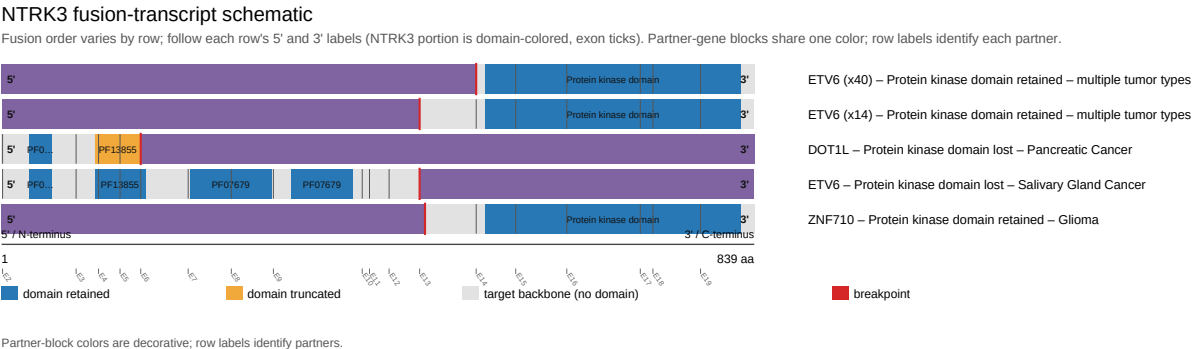
Gene	Tier	N events	In-frame%	Domain-ret. %	Fisher p	Best q
NTRK3	FDR-significant	58	98.21	96.49	0.06955	9.002e-09
ROS1	FDR-significant	122	72.73	90.68	0.214	0.000244
ETV6	FDR-significant	90	81.01	77.27	5.123e-06	0.002572
FLI1	FDR-significant	118	68.22	94.02	0.06871	0.009047
RET	Honorable mention	194	85.38	92.27	0.0004197	0.1066
FGFR2	Honorable mention	136	93.1	85.82	0.0004247	0.1066
ALK	Honorable mention	272	97.41	96.31	0.001917	0.1675
EGFR	Honorable mention	55	70	74.55	0.01145	0.1917
BRAF	Honorable mention	179	92.64	91.57	0.01337	0.1675
FGFR3	Honorable mention	152	97.37	94.04	0.01948	0.1675
CDKN2B	Honorable mention	12	66.67	20	0.02222	0.291
IKBKE	Honorable mention	12	50	36.36	0.02424	0.2663
TMPRSS2	Honorable mention	867	33.33	6.243	0.0292	0.1675
NTRK1	Honorable mention	78	72.73	84.21	0.04826	0.1675
INPPL1	Honorable mention	14	57.14	53.85	0.04895	0.3846
CDK12	Honorable mention	43	50	39.02	0.0677	0.1675
NAB2	Honorable mention	72	71.43	69.01	0.1126	0.1675
KDMSA	Honorable mention	10	50	55.56	0.119	0.7728
FH	Honorable mention	16	16.67	13.33	0.1333	0.5044

Gene highlights

NTRK3 (FDR-significant, Curated gene config)

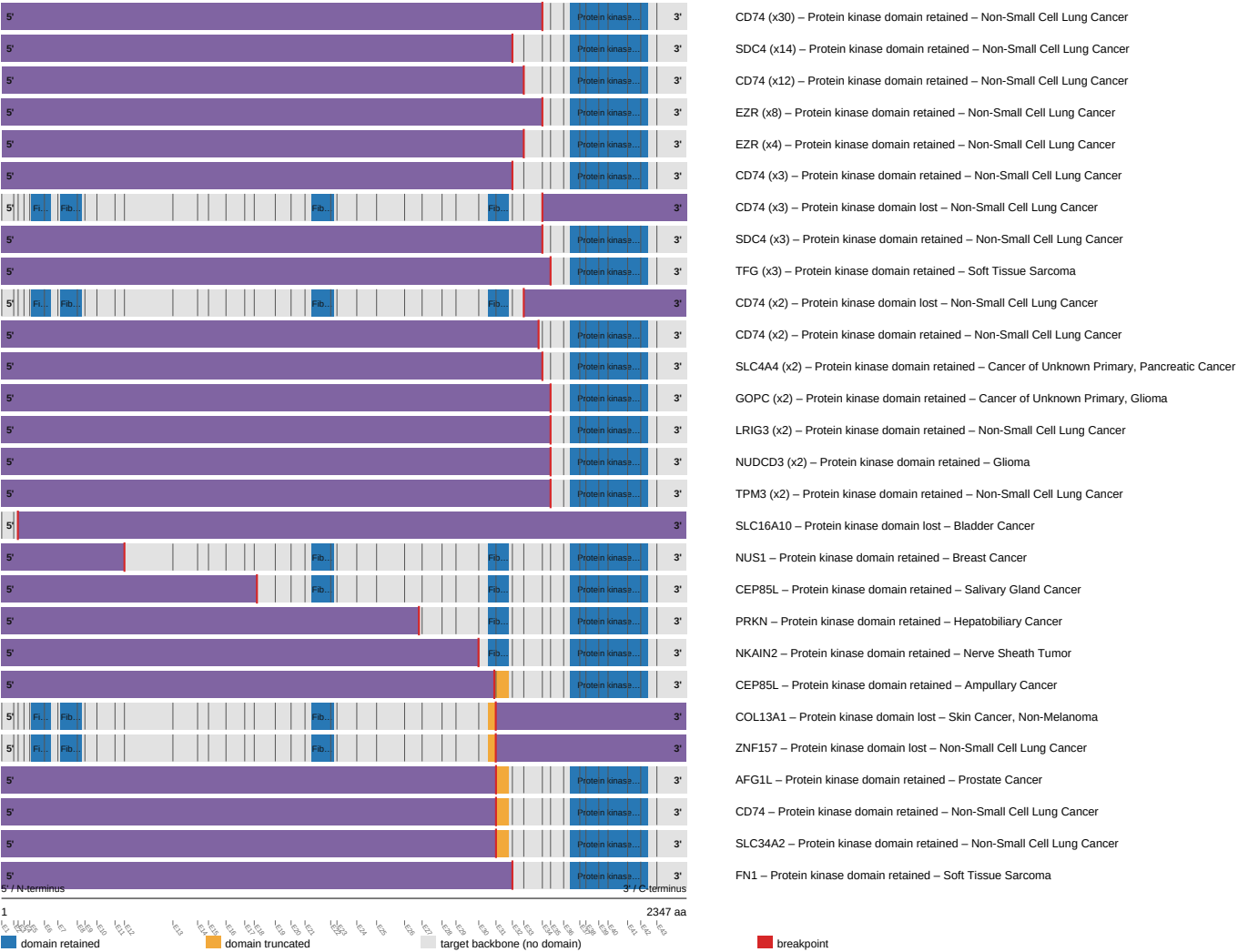
NTRK3 was analyzed across 58 fusion events, 98.2% in-frame and 96.5% domain-retained. Domain-retention Fisher's exact test on NTRK3's own fusion events alone gives $p=0.0695489$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, NTRK3's Benjamini-Hochberg-adjusted $q=9.00216e-09$ (reaches genome-wide FDR significance).

Reused from the NTRK3 individual gene report (fusion_schematic.svg).



ROS1 fusion-transcript schematic

Fusion order varies by row, follow each row's 5' and 3' labels (ROS1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/ros1/report.md](#)

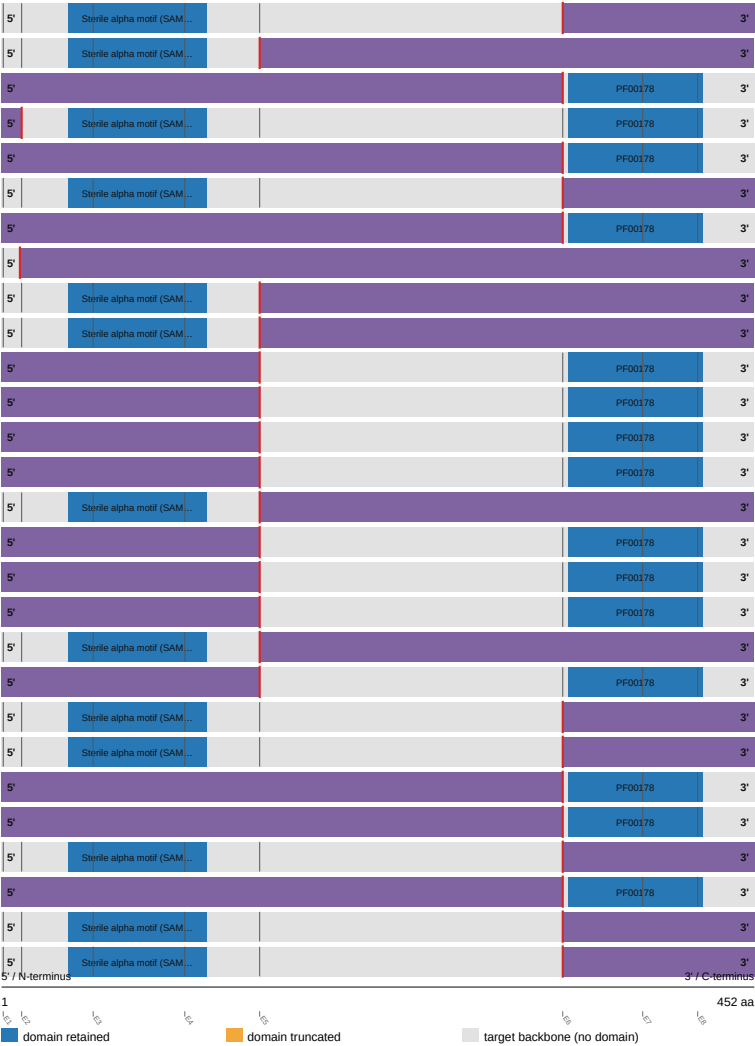
ETV6 (FDR-significant, Curated gene config)

ETV6 was analyzed across 90 fusion events, 81.0% in-frame and 77.3% domain-retained. Domain-retention Fisher's exact test on ETV6's own fusion events alone gives $p=5.12326e-06$ (statistically significant at $\alpha=0.05$). The Sterile alpha motif (SAM)/Pointed domain appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, ETV6's Benjamini-Hochberg-adjusted $q=0.00257187$ (reaches genome-wide FDR significance).

Reused from the ETV6 individual gene report ([fusion_schematic.svg](#)).

ETV6 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (ETV6 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 30 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

- NTRK3 (x43) – Sterile alpha motif (SAM)/Pointed domain retained – multiple tumor types
- NTRK3 (x11) – Sterile alpha motif (SAM)/Pointed domain retained – Thyroid Cancer, Pancreatic Cancer
- BCL2L14 (x3) – Sterile alpha motif (SAM)/Pointed domain lost – Breast Cancer, Esophagogastric Cancer
- HCFC1 (x2) – Sterile alpha motif (SAM)/Pointed domain retained – Breast Cancer
- BORCS5 (x2) – Sterile alpha motif (SAM)/Pointed domain lost – Breast Cancer
- SGO2 (x2) – Sterile alpha motif (SAM)/Pointed domain retained – Bladder Cancer
- SLIT2 (x2) – Sterile alpha motif (SAM)/Pointed domain lost – Ovarian Cancer
- CDKN1B – Sterile alpha motif (SAM)/Pointed domain lost – Head and Neck Cancer
- ABCC9 – Sterile alpha motif (SAM)/Pointed domain retained – Melanoma
- ANKS1B – Sterile alpha motif (SAM)/Pointed domain retained – Breast Cancer
- ARHGAP26 – Sterile alpha motif (SAM)/Pointed domain lost – Cervical Cancer
- BORCS5 – Sterile alpha motif (SAM)/Pointed domain lost – Prostate Cancer
- CUX2 – Sterile alpha motif (SAM)/Pointed domain lost – Breast Cancer
- GALNT18 – Sterile alpha motif (SAM)/Pointed domain lost – Breast Cancer
- GNB1 – Sterile alpha motif (SAM)/Pointed domain retained – Esophagogastric Cancer
- GRIN2B – Sterile alpha motif (SAM)/Pointed domain lost – Bladder Cancer
- NOL4 – Sterile alpha motif (SAM)/Pointed domain lost – Breast Cancer
- NTRK3 – Sterile alpha motif (SAM)/Pointed domain lost – Salivary Gland Cancer
- PLEKHG7 – Sterile alpha motif (SAM)/Pointed domain retained – Breast Cancer
- PTPRN2 – Sterile alpha motif (SAM)/Pointed domain lost – Peritoneal Cancer, NOS
- AEBP2 – Sterile alpha motif (SAM)/Pointed domain retained – Prostate Cancer
- APOLD1 – Sterile alpha motif (SAM)/Pointed domain retained – Non-Small Cell Lung Cancer
- FAM234B – Sterile alpha motif (SAM)/Pointed domain lost – Non-Small Cell Lung Cancer
- FGFR1OP2 – Sterile alpha motif (SAM)/Pointed domain lost – Melanoma
- IKZF3 – Sterile alpha motif (SAM)/Pointed domain retained – Cancer of Unknown Primary
- LRP6 – Sterile alpha motif (SAM)/Pointed domain lost – Glioma
- PDE3A – Sterile alpha motif (SAM)/Pointed domain retained – Esophagogastric Cancer
- SUFU – Sterile alpha motif (SAM)/Pointed domain retained – Non-Small Cell Lung Cancer

Full per-gene detail: [gene_reports/etv6/report.md](#)

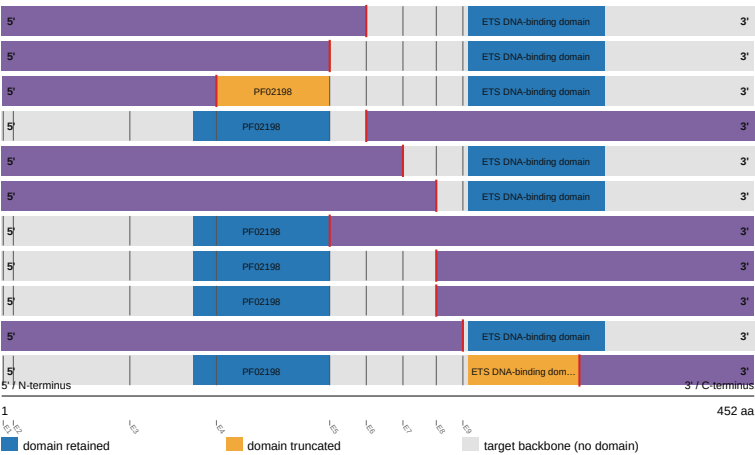
FLI1 (FDR-significant, Curated gene config)

FLI1 was analyzed across 118 fusion events, 68.2% in-frame and 94.0% domain-retained. Domain-retention Fisher's exact test on FLI1's own fusion events alone gives $p=0.0687101$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, FLI1's Benjamini-Hochberg-adjusted $q=0.00904727$ (reaches genome-wide FDR significance).

Reused from the FLI1 individual gene report ([fusion_schematic.svg](#)).

FLI1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FLI1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



- EWSR1 (x76) – ETS DNA-binding domain retained – multiple tumor types
- EWSR1 (x26) – ETS DNA-binding domain retained – Bone Cancer, Salivary Gland Cancer, Soft Tissue
- EWSR1 (x3) – ETS DNA-binding domain retained – Bone Cancer
- EWSR1 (x3) – ETS DNA-binding domain lost – Bone Cancer
- EWSR1 (x2) – ETS DNA-binding domain retained – Bone Cancer
- EWSR1 (x2) – ETS DNA-binding domain retained – Bone Cancer
- EWSR1 – ETS DNA-binding domain lost – Bone Cancer
- ATM – ETS DNA-binding domain lost – Prostate Cancer
- EWSR1 – ETS DNA-binding domain lost – Bone Cancer
- EWSR1 – ETS DNA-binding domain retained – Bone Cancer
- ERG – ETS DNA-binding domain truncated – Esophagogastric Cancer

Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: [gene_reports/fli1/report.md](#)

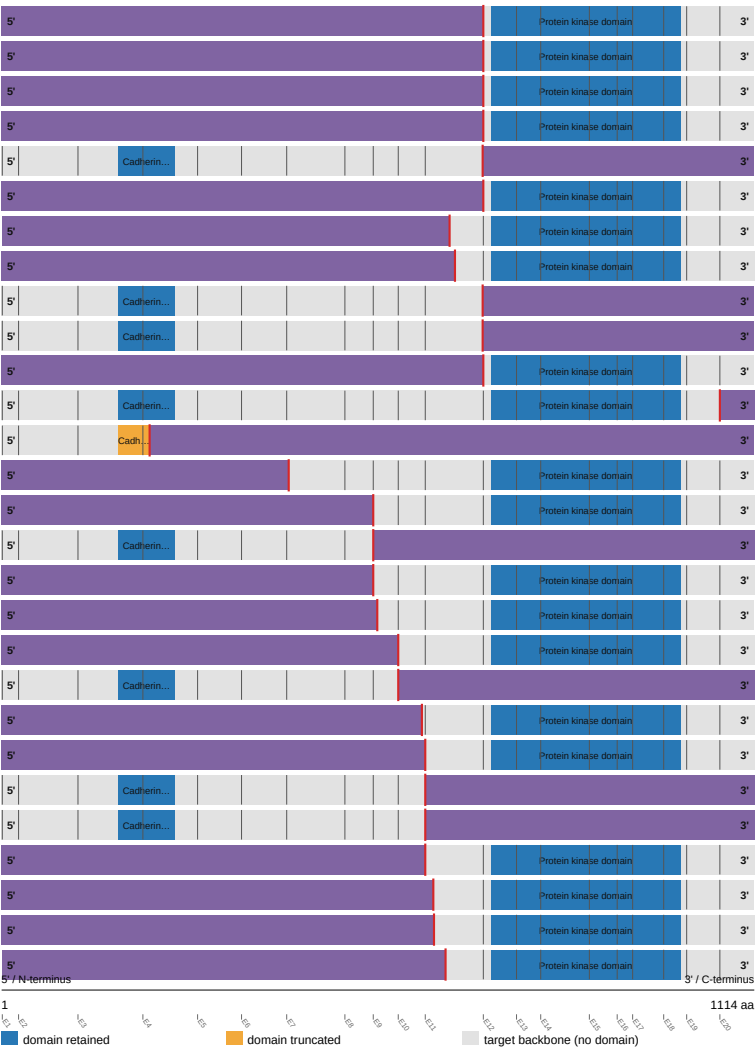
RET (Honorable mention, Curated gene config)

RET was analyzed across 194 fusion events, 85.4% in-frame and 92.3% domain-retained. Domain-retention Fisher's exact test on RET's own fusion events alone gives $p=0.000419666$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, RET's Benjamini-Hochberg-adjusted $q=0.106596$ (does not reach genome-wide FDR significance). The Cadherin domain appears to require loss or disruption rather than retention. Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RET individual gene report ([fusion_schematic.svg](#)).

RET fusion-transcript schematic

Fusion order varies by row, follow each row's 5' and 3' labels (RET portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



- KIF5B (x78) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Cancer of Unknown Primary
- CCDC6 (x45) – Protein kinase domain retained – multiple tumor types
- NCOA4 (x16) – Protein kinase domain retained – multiple tumor types
- SPECC1L (x4) – Protein kinase domain retained – Soft Tissue Sarcoma, Breast Cancer, Renal Cell Carcinoma
- ERCC6 (x3) – Protein kinase domain lost – Non-Small Cell Lung Cancer
- TIMM23B (x3) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Thyroid Cancer
- RUFY1 (x2) – Protein kinase domain retained – Thyroid Cancer
- KIF5B (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
- CCDC6 (x2) – Protein kinase domain lost – Thyroid Cancer
- KIF5B (x2) – Protein kinase domain lost – Non-Small Cell Lung Cancer
- RUFY3 (x2) – Protein kinase domain retained – Thyroid Cancer
- FBXL7 (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
- UXS1 – Protein kinase domain lost – Hepatobiliary Cancer
- KIAA1217 – Protein kinase domain retained – Esophagogastric Cancer
- KCNMA1 – Protein kinase domain retained – Esophagogastric Cancer
- KIF5B – Protein kinase domain lost – Non-Small Cell Lung Cancer
- KIF5B – Protein kinase domain retained – Non-Small Cell Lung Cancer
- DOCK1 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- CCDC6 – Protein kinase domain retained – Esophagogastric Cancer
- LIN52 – Protein kinase domain lost – Prostate Cancer
- KIF5B – Protein kinase domain retained – Non-Small Cell Lung Cancer
- CCDC6 – Protein kinase domain retained – Colorectal Cancer
- CCDC6 – Protein kinase domain lost – Thyroid Cancer
- PDCD10 – Protein kinase domain lost – Thyroid Cancer
- TFG – Protein kinase domain retained – Thyroid Cancer
- ERC1 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- CCDC6 – Protein kinase domain retained – Thyroid Cancer
- RBPM5 – Protein kinase domain retained – Non-Small Cell Lung Cancer

Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 45 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/ret/report.md](#)

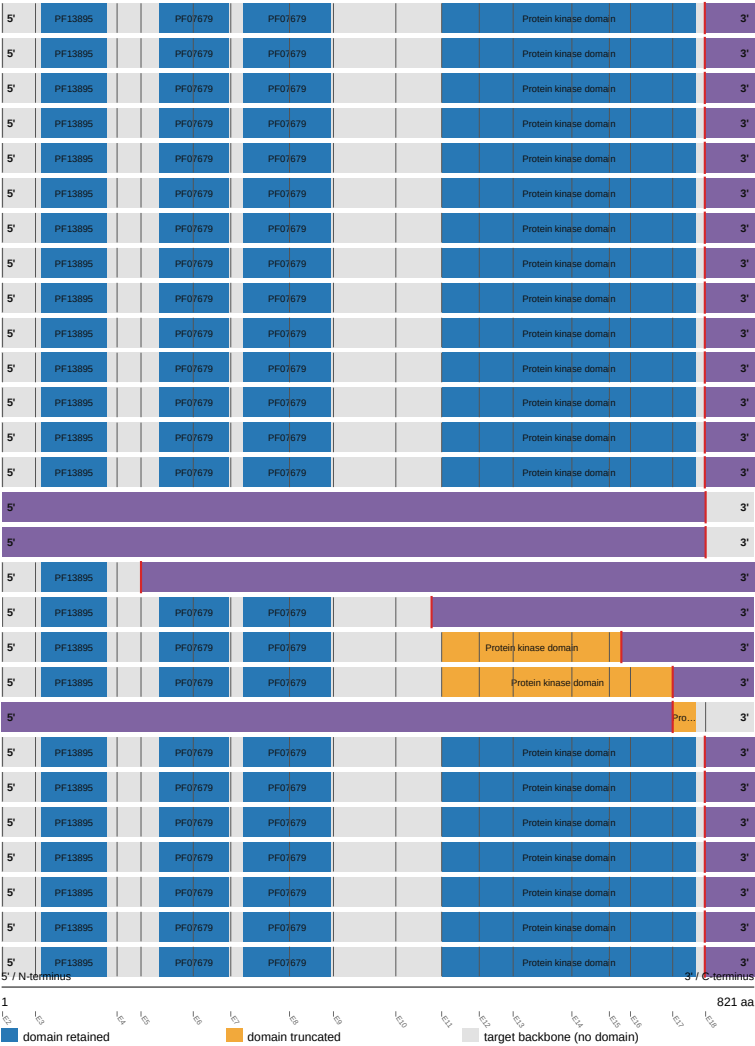
FGFR2 (Honorable mention, Curated gene config)

FGFR2 was analyzed across 136 fusion events, 93.1% in-frame and 85.8% domain-retained. Domain-retention Fisher's exact test on FGFR2's own fusion events alone gives $p=0.000424687$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, FGFR2's Benjamini-Hochberg-adjusted $q=0.106596$ (does not reach genome-wide FDR significance). The C-terminal autoinhibitory tail (exon 18) appears to require loss or disruption rather than retention. Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR2 individual gene report ([fusion_schematic.svg](#)).

FGFR2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 84 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/fgfr2/report.md](#)

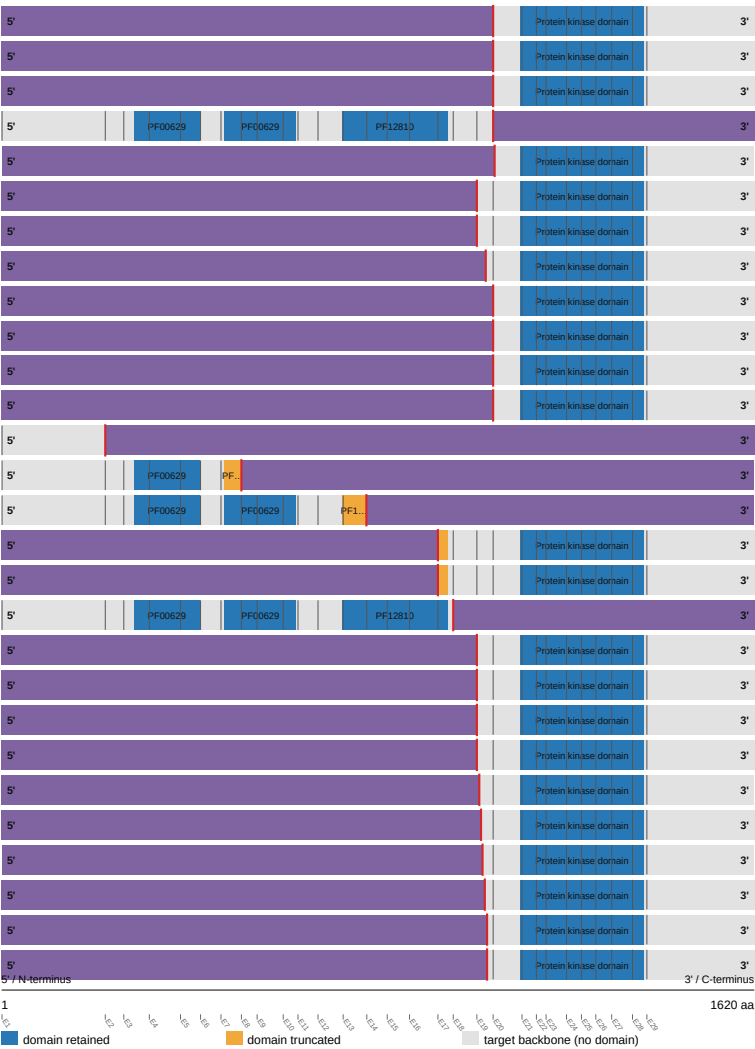
ALK (Honorable mention, Curated gene config)

ALK was analyzed across 272 fusion events, 97.4% in-frame and 96.3% domain-retained. Domain-retention Fisher's exact test on ALK's own fusion events alone gives $p=0.00191661$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, ALK's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the ALK individual gene report ([fusion_schematic.svg](#)).

ALK fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (ALK portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



- EML4 (x204) – Protein kinase domain retained – multiple tumor types
- STRN (x5) – Protein kinase domain retained – Pancreatic Cancer, Non-Small Cell Lung Cancer
- KIF5B (x4) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Soft Tissue Sarcoma
- EML4 (x3) – Protein kinase domain lost – Non-Small Cell Lung Cancer
- PICALM (x3) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Cancer of Unknown Primary
- CLTC (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer, Soft Tissue Sarcoma
- EML4 (x2) – Protein kinase domain retained – Cancer of Unknown Primary, Non-Small Cell Lung Cancer
- EML4 (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
- RANBP2 (x2) – Protein kinase domain retained – Soft Tissue Sarcoma
- SPTBN1 (x2) – Protein kinase domain retained – Colorectal Cancer
- SQSTM1 (x2) – Protein kinase domain retained – Non-Small Cell Lung Cancer
- TPM1 (x2) – Protein kinase domain retained – Mesothelioma
- ASAP2 – Protein kinase domain lost – Pancreatic Cancer
- CTSE – Protein kinase domain lost – Prostate Cancer
- H6PD – Protein kinase domain lost – Uterine Sarcoma
- EMILIN1 – Protein kinase domain retained – Melanoma
- EML4 – Protein kinase domain retained – Pancreatic Cancer
- LRP1B – Protein kinase domain lost – Bladder Cancer
- FN1 – Protein kinase domain retained – Soft Tissue Sarcoma
- KIF5B – Protein kinase domain retained – Non-Small Cell Lung Cancer
- NOTCH1 – Protein kinase domain retained – Glioma
- NRP2 – Protein kinase domain retained – Uterine Sarcoma
- EML4 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- EML4 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- PLEKHA7 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- EML4 – Protein kinase domain retained – Colorectal Cancer
- EML4 – Protein kinase domain retained – Non-Small Cell Lung Cancer
- STRN – Protein kinase domain retained – Colorectal Cancer

Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/alk/report.md](#)

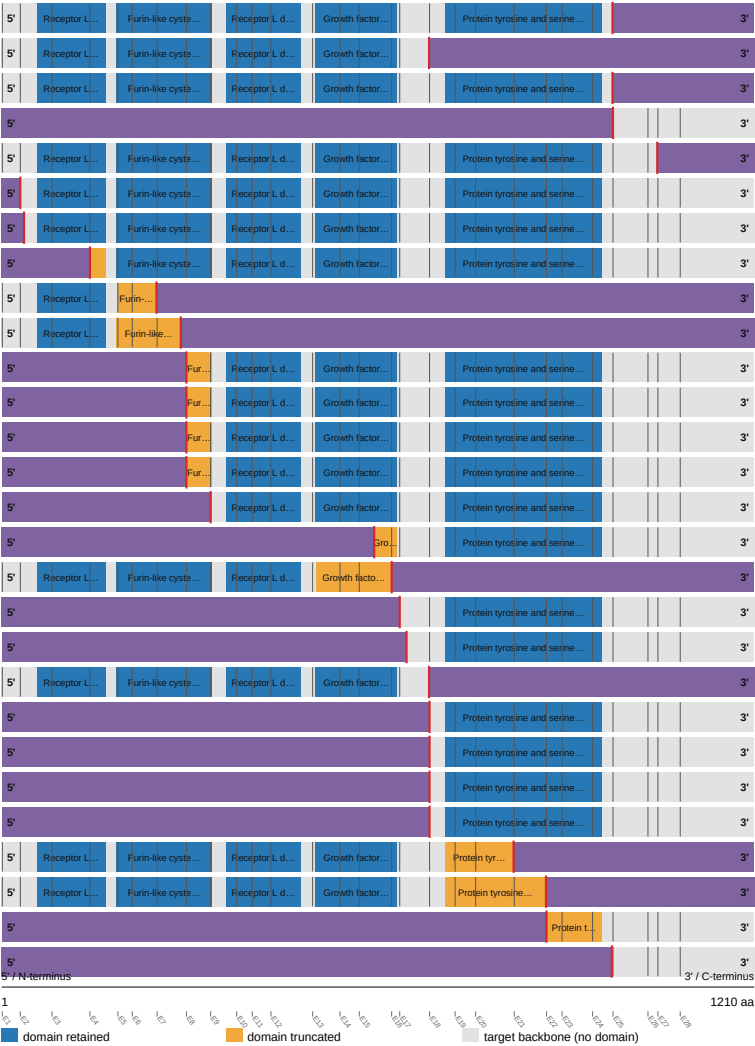
EGFR (Honorable mention)

EGFR was analyzed across 55 fusion events, 70.0% in-frame and 74.5% domain-retained. Domain-retention Fisher's exact test on EGFR's own fusion events alone gives $p=0.0114539$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, EGFR's Benjamini-Hochberg-adjusted $q=0.191661$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the EGFR individual gene report ([fusion_schematic.svg](#)).

EGFR fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (EGFR portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

- SEPTIN14 (x12) – Protein tyrosine and serine/threonine kinase retained – Glioma, Ampullary Cancer, E
- FADD (x2) – Protein tyrosine and serine/threonine kinase lost – Non-Small Cell Lung Cancer, Small Cell
- RAD51 (x2) – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer
- NIPSNAP2 (x2) – Protein tyrosine and serine/threonine kinase lost – Breast Cancer
- SEPTIN14 (x2) – Protein tyrosine and serine/threonine kinase retained – Glioma
- PLOD3 – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer
- ZBPB – Protein tyrosine and serine/threonine kinase retained – Glioma
- SCAF4 – Protein tyrosine and serine/threonine kinase retained – Bone Cancer
- VOPPI – Protein tyrosine and serine/threonine kinase lost – Breast Cancer
- PCDH15 – Protein tyrosine and serine/threonine kinase lost – Renal Cell Carcinoma
- DDC – Protein tyrosine and serine/threonine kinase retained – Breast Cancer
- GARS1 – Protein tyrosine and serine/threonine kinase retained – Glioma
- SEL1L – Protein tyrosine and serine/threonine kinase retained – Pancreatic Cancer
- TNS3 – Protein tyrosine and serine/threonine kinase retained – Glioma
- CSF2RA – Protein tyrosine and serine/threonine kinase retained – Melanoma
- NIPSNAP2 – Protein tyrosine and serine/threonine kinase retained – Glioma
- TUT7 – Protein tyrosine and serine/threonine kinase lost – Non-Small Cell Lung Cancer
- YIF1B – Protein tyrosine and serine/threonine kinase retained – Soft Tissue Sarcoma
- PDE1C – Protein tyrosine and serine/threonine kinase retained – Glioma
- VSTM2A – Protein tyrosine and serine/threonine kinase lost – Glioma
- CADPS – Protein tyrosine and serine/threonine kinase retained – Germ Cell Tumor
- EEA1 – Protein tyrosine and serine/threonine kinase retained – Colorectal Cancer
- KIF5B – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer
- NUMA1 – Protein tyrosine and serine/threonine kinase retained – Colorectal Cancer
- CDH7 – Protein tyrosine and serine/threonine kinase truncated – Bladder Cancer
- LANCL2 – Protein tyrosine and serine/threonine kinase truncated – Glioma
- VWC2 – Protein tyrosine and serine/threonine kinase truncated – Glioma
- ELAPOR2 – Protein tyrosine and serine/threonine kinase lost – Glioma

Full per-gene detail: [gene_reports/egfr/report.md](#)

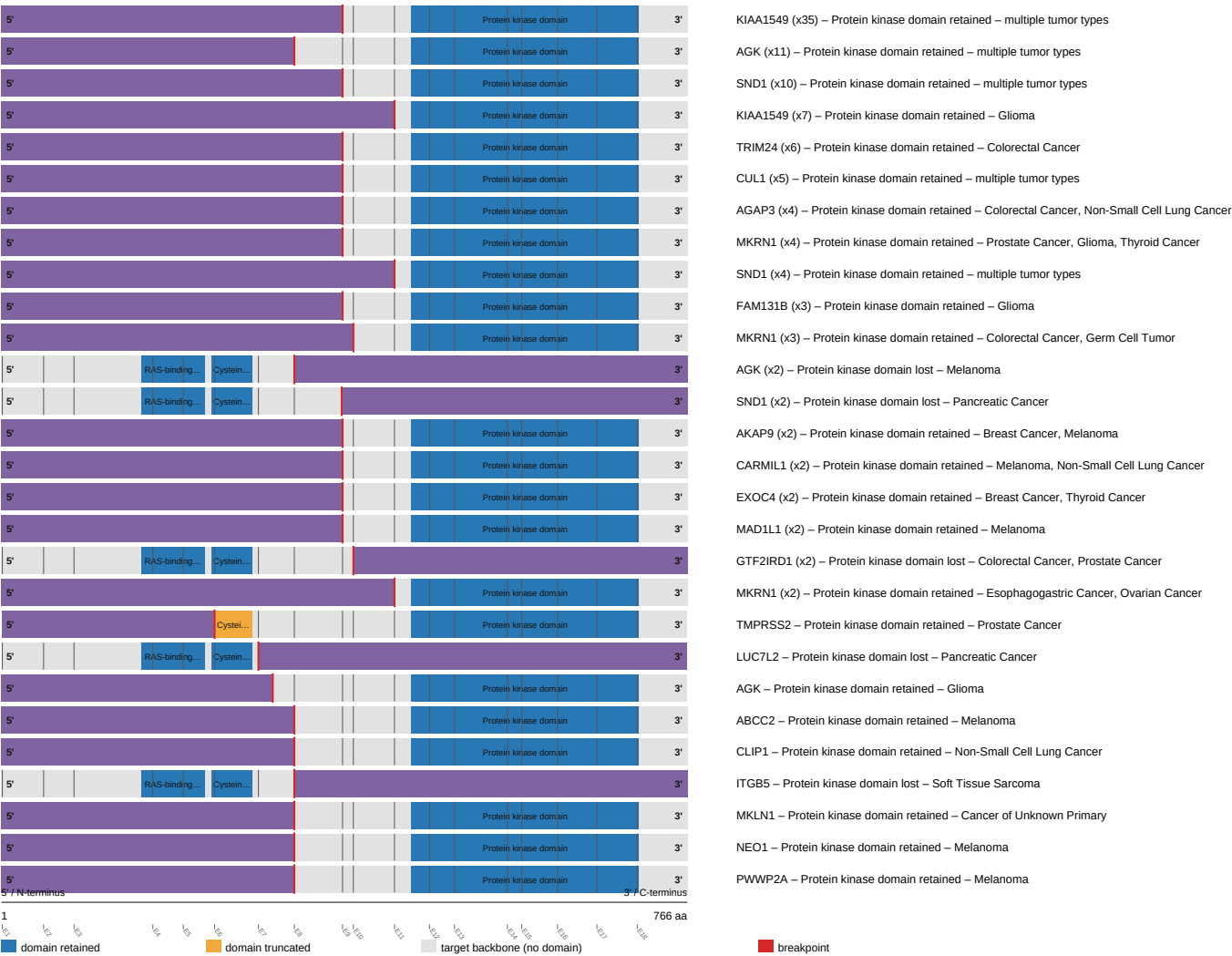
BRAF (Honorable mention, Curated gene config)

BRAF was analyzed across 179 fusion events, 92.6% in-frame and 91.6% domain-retained. Domain-retention Fisher's exact test on BRAF's own fusion events alone gives $p=0.0133676$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, BRAF's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the BRAF individual gene report ([fusion_schematic.svg](#)).

BRAF fusion-transcript schematic

Fusion order varies by row, follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 89 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/braf/report.md](#)

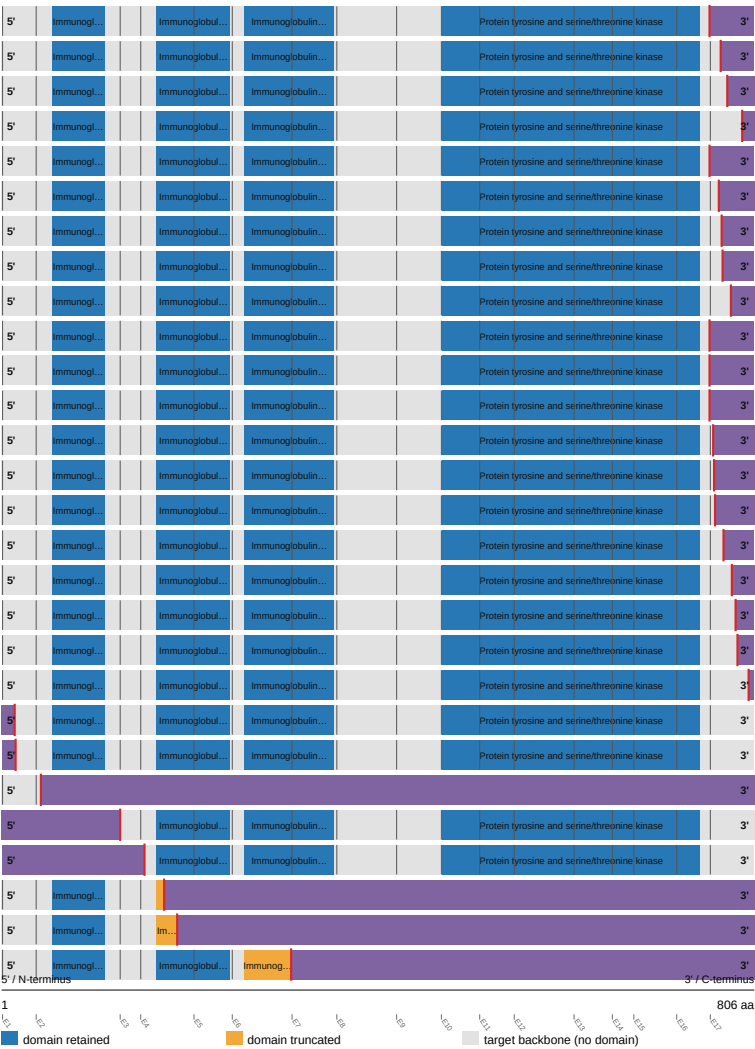
FGFR3 (Honorable mention)

FGFR3 was analyzed across 152 fusion events, 97.4% in-frame and 94.0% domain-retained. Domain-retention Fisher's exact test on FGFR3's own fusion events alone gives $p=0.0194824$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. Among 359 genes scanned genome-wide in this cohort run, FGFR3's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR3 individual gene report ([fusion_schematic.svg](#)).

FGFR3 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR3 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 51 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

- TACC3 (x71) – Protein tyrosine and serine/threonine kinase retained – multiple tumor types
- TACC3 (x4) – Protein tyrosine and serine/threonine kinase retained – multiple tumor types
- TACC3 (x4) – Protein tyrosine and serine/threonine kinase retained – Glioma, Miscellaneous Neuroepithelial Tumors
- TACC3 (x4) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- TNIP2 (x3) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- TACC3 (x3) – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer
- TACC3 (x3) – Protein tyrosine and serine/threonine kinase retained – Glioma, Bladder Cancer
- TACC3 (x3) – Protein tyrosine and serine/threonine kinase retained – Glioma, Bladder Cancer
- TACC3 (x3) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- CCDC149 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- JAKMIP1 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer, Colorectal Cancer
- LPCAT2 (x2) – Protein tyrosine and serine/threonine kinase retained – Hepatobiliary Cancer
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Esophagogastric Cancer, Glioma
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Glioma
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer, Glioma
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Bladder Cancer, Endometrial Cancer
- TACC3 (x2) – Protein tyrosine and serine/threonine kinase retained – Glioma
- DGKQ – Protein tyrosine and serine/threonine kinase retained – Endometrial Cancer
- GRK4 – Protein tyrosine and serine/threonine kinase retained – Breast Cancer
- TACC3 – Protein tyrosine and serine/threonine kinase lost – Ovarian Cancer
- NSD2 – Protein tyrosine and serine/threonine kinase retained – Endometrial Cancer
- BABAM2 – Protein tyrosine and serine/threonine kinase retained – Endometrial Cancer
- RBM20 – Protein tyrosine and serine/threonine kinase lost – Endometrial Cancer
- TACC3 – Protein tyrosine and serine/threonine kinase lost – Glioma
- TACC3 – Protein tyrosine and serine/threonine kinase lost – Non-Small Cell Lung Cancer

Full per-gene detail: [gene_reports/fgfr3/report.md](#)

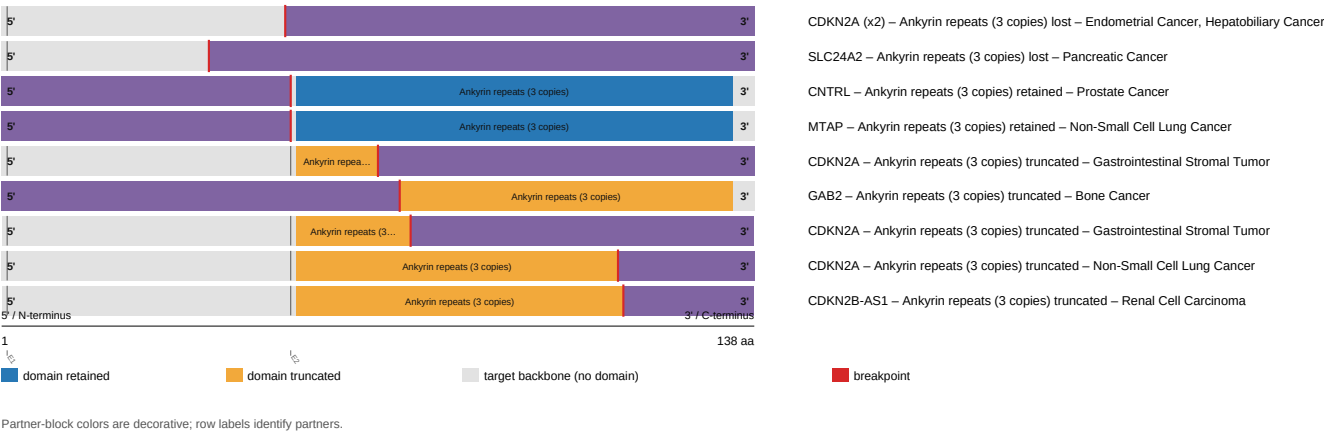
CDKN2B (Honorable mention)

CDKN2B was analyzed across 12 fusion events, 66.7% in-frame and 20.0% domain-retained. Domain-retention Fisher's exact test on CDKN2B's own fusion events alone gives $p=0.0222222$ (statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, CDKN2B's Benjamini-Hochberg-adjusted $q=0.291014$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDKN2B individual gene report ([fusion_schematic.svg](#)).

CDKN2B fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (CDKN2B portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Full per-gene detail: [gene_reports/cdkn2b/report.md](#)

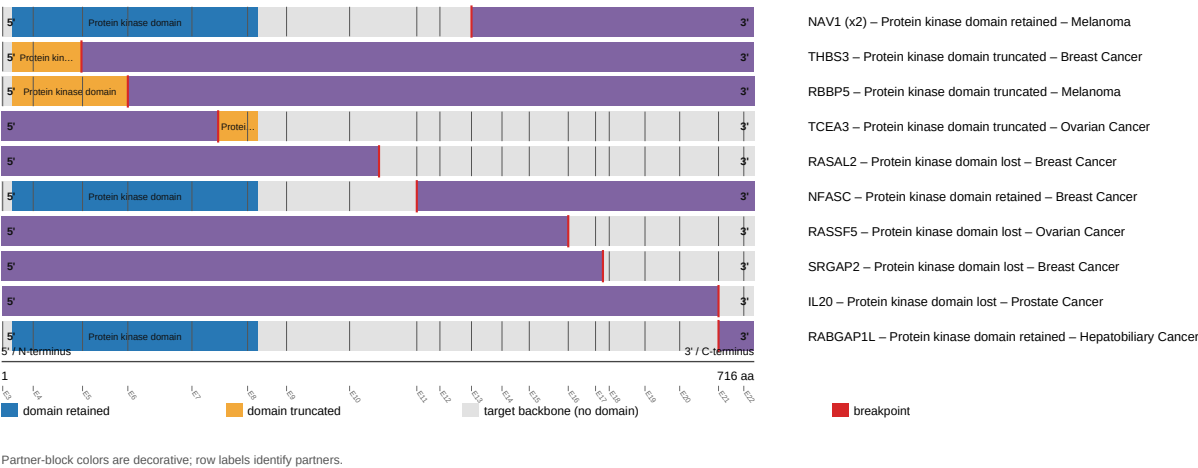
IKBKE (Honorable mention)

IKBKE was analyzed across 12 fusion events, 50.0% in-frame and 36.4% domain-retained. Domain-retention Fisher's exact test on IKBKE's own fusion events alone gives $p=0.0242424$ (statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, IKBKE's Benjamini-Hochberg-adjusted $q=0.266266$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the IKBKE individual gene report ([fusion_schematic.svg](#)).

IKBKE fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (IKBKE portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Full per-gene detail: [gene_reports/ikbke/report.md](#)

TMPRSS2 (Honorable mention)

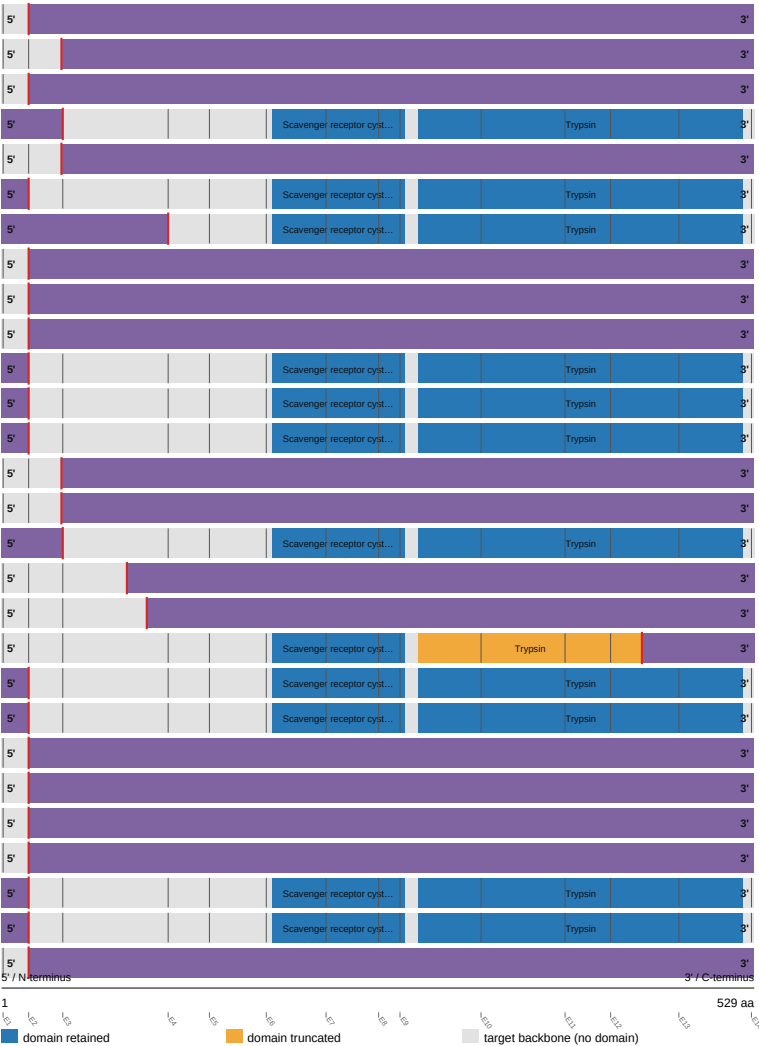
TMPRSS2 was analyzed across 867 fusion events, 33.3% in-frame and 6.2% domain-retained. Domain-retention Fisher's exact test on TMPRSS2's own fusion events alone gives $p=0.0292045$ (statistically significant at $\alpha=0.05$). The Trypsin appears to be required for retention. Among 359 genes scanned

genome-wide in this cohort run, TMPRSS2's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the TMPRSS2 individual gene report (*fusion_schematic.svg*).

TMPRSS2 fusion-transcript schematic

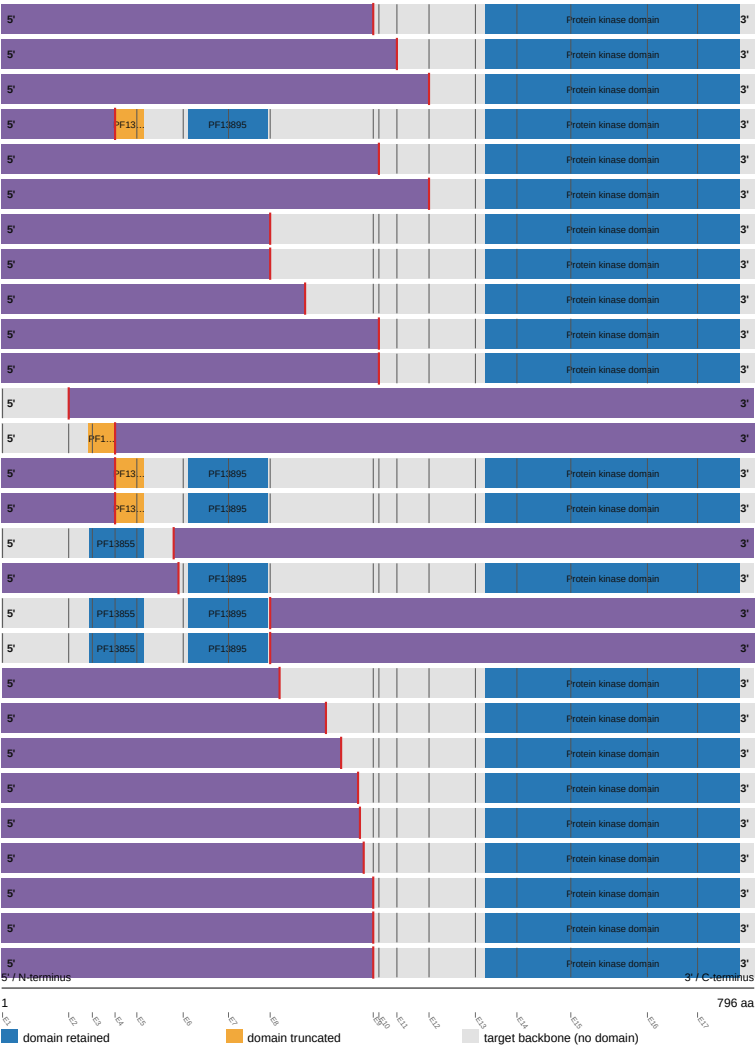
Fusion order varies by row; follow each row's 5' and 3' labels (TMPRSS2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Reused from the NTRK1 individual gene report (fusion_schematic.svg).

NTRK1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NTRK1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

- TPM3 (x5) – Protein kinase domain retained – Thyroid Cancer, Colorectal Cancer, Glioma
- LMNA (x5) – Protein kinase domain retained – Breast Cancer, Colorectal Cancer, Soft Tissue Sarcoma
- LMNA (x5) – Protein kinase domain retained – Hepatobiliary Cancer, Colorectal Cancer
- SLAMF6 (x3) – Protein kinase domain retained – Bladder Cancer
- PLEKHA6 (x3) – Protein kinase domain retained – Hepatobiliary Cancer
- TPM3 (x3) – Protein kinase domain retained – Soft Tissue Sarcoma, Thyroid Cancer
- BCAN (x2) – Protein kinase domain retained – Glioma, Soft Tissue Sarcoma
- CTRC (x2) – Protein kinase domain retained – Pancreatic Cancer
- TPM3 (x2) – Protein kinase domain retained – Colorectal Cancer
- IRF2BP2 (x2) – Protein kinase domain retained – Cancer of Unknown Primary, Non-Small Cell Lung Ca
- TPR (x2) – Protein kinase domain retained – Soft Tissue Sarcoma, Thyroid Cancer
- SHPRH – Protein kinase domain lost – Soft Tissue Sarcoma
- CADM3 – Protein kinase domain lost – Non-Small Cell Lung Cancer
- PRCC – Protein kinase domain retained – Small Cell Lung Cancer
- TAF2A – Protein kinase domain retained – Soft Tissue Sarcoma
- P2RY8 – Protein kinase domain lost – Non-Small Cell Lung Cancer
- TARS2 – Protein kinase domain retained – Soft Tissue Sarcoma
- NELFCD – Protein kinase domain lost – Ovarian Cancer
- SEMA4A – Protein kinase domain lost – Prostate Cancer
- NOS1AP – Protein kinase domain retained – Esophagogastric Cancer
- TPM3 – Protein kinase domain retained – Thyroid Cancer
- TPR – Protein kinase domain retained – Colorectal Cancer
- TPR – Protein kinase domain retained – Thyroid Cancer
- GON4L – Protein kinase domain retained – Melanoma
- KIF21B – Protein kinase domain retained – Miscellaneous Brain Tumor
- AFAP1 – Protein kinase domain retained – Glioma
- DIAPH1 – Protein kinase domain retained – Thyroid Cancer
- IRF2BP2 – Protein kinase domain retained – Thyroid Cancer

Full per-gene detail: [gene_reports/ntrk1/report.md](#)

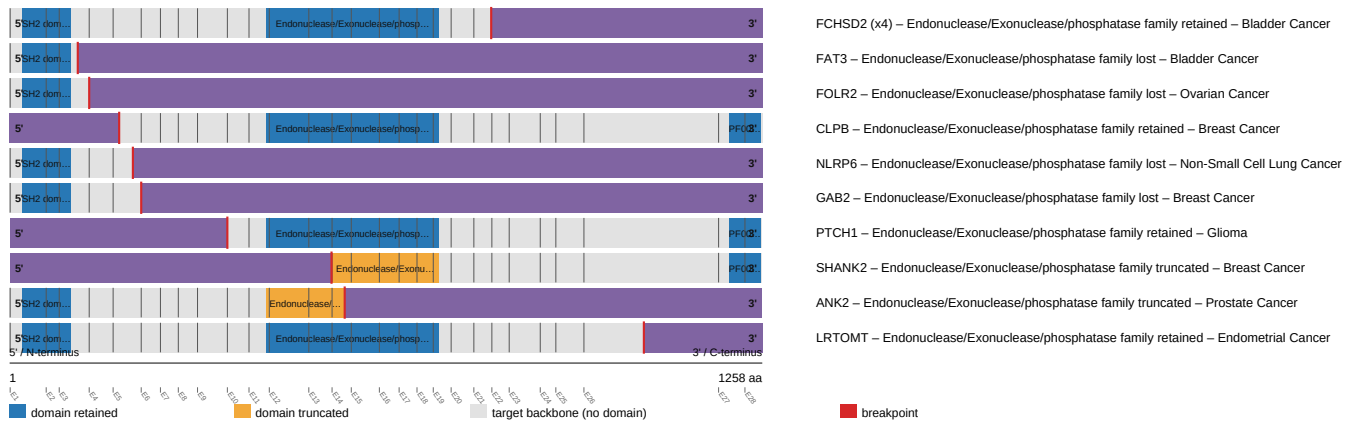
INPPL1 (Honorable mention)

INPPL1 was analyzed across 14 fusion events, 57.1% in-frame and 53.8% domain-retained. Domain-retention Fisher's exact test on INPPL1's own fusion events alone gives $p=0.048951$ (statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, INPPL1's Benjamini-Hochberg-adjusted $q=0.384608$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the INPPL1 individual gene report (fusion_schematic.svg).

INPPL1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (INPPL1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: gene_reports/inppl1/report.md

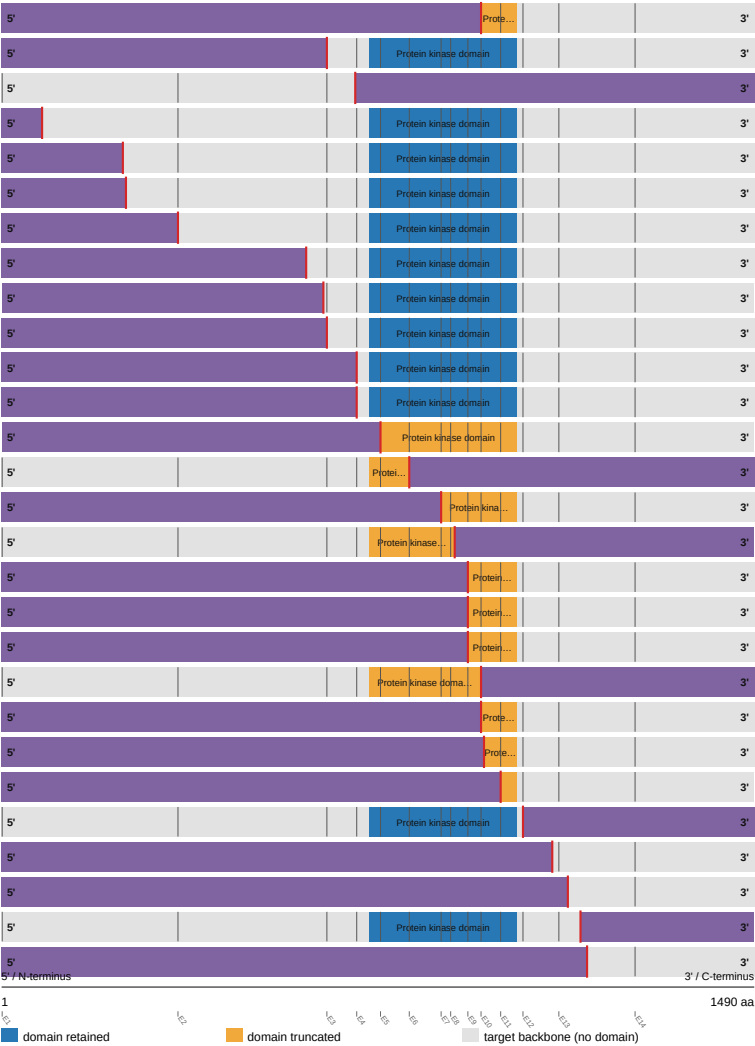
CDK12 (Honorable mention)

CDK12 was analyzed across 43 fusion events, 50.0% in-frame and 39.0% domain-retained. Domain-retention Fisher's exact test on CDK12's own fusion events alone gives $p=0.0677006$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, CDK12's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDK12 individual gene report (fusion_schematic.svg).

CDK12 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (CDK12 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



- STAT5B (x4) – Protein kinase domain truncated – Salivary Gland Cancer
- ADCY9 (x2) – Protein kinase domain retained – Breast Cancer
- HTR1E (x2) – Protein kinase domain lost – Prostate Cancer
- VMP1 – Protein kinase domain retained – Breast Cancer
- IKZF3 – Protein kinase domain retained – Cancer of Unknown Primary
- FBXL20 – Protein kinase domain retained – Colorectal Cancer
- ANKFN1 – Protein kinase domain retained – Breast Cancer
- MED13 – Protein kinase domain retained – Breast Cancer
- AKAP8L – Protein kinase domain retained – Breast Cancer
- ERBB2 – Protein kinase domain retained – Hepatobiliary Cancer
- ODAD4 – Protein kinase domain retained – Esophagogastric Cancer
- TBKBP1 – Protein kinase domain retained – Breast Cancer
- GRB7 – Protein kinase domain truncated – Endometrial Cancer
- RECQL4 – Protein kinase domain truncated – Endometrial Cancer
- COLEC12 – Protein kinase domain truncated – Colorectal Cancer
- CACNA1G – Protein kinase domain truncated – Breast Cancer
- ERBB2 – Protein kinase domain truncated – Ovarian Cancer
- FBXL20 – Protein kinase domain truncated – Cancer of Unknown Primary
- SYNRG – Protein kinase domain truncated – Cancer of Unknown Primary
- SKAP1 – Protein kinase domain truncated – Breast Cancer
- USP6 – Protein kinase domain truncated – Breast Cancer
- ERBB2 – Protein kinase domain truncated – Ovarian Cancer
- CSF3 – Protein kinase domain truncated – Ovarian Cancer
- FBXL20 – Protein kinase domain retained – Colorectal Cancer
- FAM222B – Protein kinase domain lost – Bladder Cancer
- FBXL20 – Protein kinase domain lost – Prostate Cancer
- SPON1 – Protein kinase domain retained – Breast Cancer
- FBXL20 – Protein kinase domain lost – Head and Neck Cancer

Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: [gene_reports/cdk12/report.md](#)

NAB2 (Honorable mention)

NAB2 was analyzed across 72 fusion events, 71.4% in-frame and 69.0% domain-retained. Domain-retention Fisher's exact test on NAB2's own fusion events alone gives $p=0.11258$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, NAB2's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NAB2 individual gene report (fusion_schematic.svg).

NAB2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NAB2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Full per-gene detail: [gene_reports/nab2/report.md](#)

KDM5A (Honorable mention)

KDM5A was analyzed across 10 fusion events, 50.0% in-frame and 55.6% domain-retained. Domain-retention Fisher's exact test on KDM5A's own fusion events alone gives $p=0.119048$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, KDM5A's Benjamini-Hochberg-adjusted $q=0.772783$ (does not reach genome-wide FDR significance). Did not survive genome-wide multiple-testing correction (FDR-adjusted q -value at or above the significance threshold), but ranks highly by raw p -value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the KDM5A individual gene report ([fusion_schematic.svg](#)).

Fusion order varies by row; follow each row's 5' and 3' labels (KDM5A portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



FH (Honorable mention)

Reused from the FH individual gene report (fusion_schematic.svg).

Fusion order varies by row; follow each row's 5' and 3' labels (FH portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



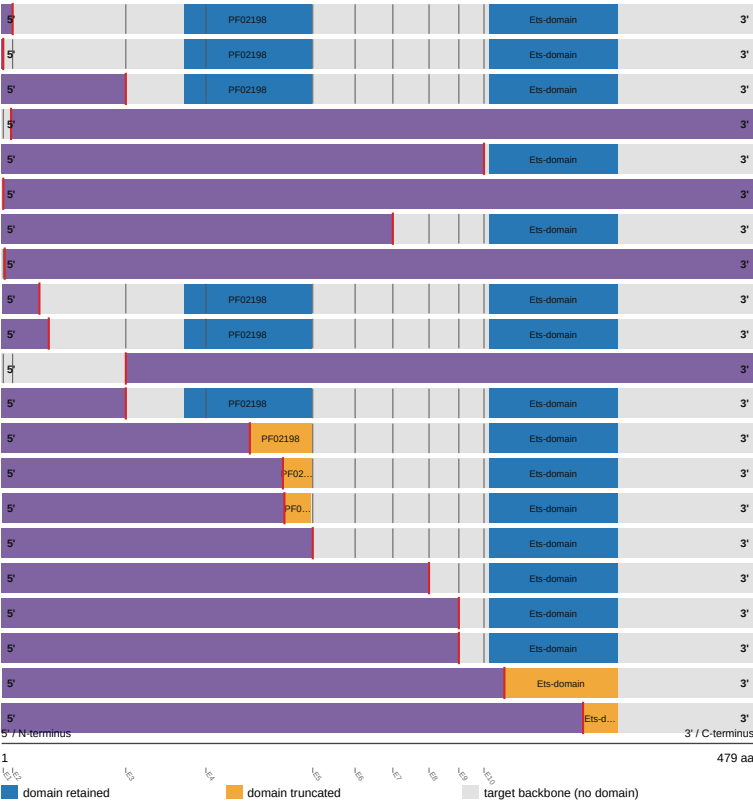
ERG (Curated gene config)

ERG was analyzed across 788 fusion events, 33.0% in-frame and 97.7% domain-retained. Domain-retention Fisher's exact test on ERG's own fusion events alone gives $p=0.846156$ (not statistically significant at $\alpha=0.05$). Among 359 genes scanned genome-wide in this cohort run, ERG's Benjamini-Hochberg-adjusted $q=0.167538$ (does not reach genome-wide FDR significance).

Reused from the ERG individual gene report (*fusion_schematic.svg*).

ERG fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (ERG portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



TMPRSS2 (x372) – Ets-domain retained – Prostate Cancer, Cancer of Unknown Primary, Gastrointestri
TMPRSS2 (x336) – Ets-domain retained – Prostate Cancer, Cancer of Unknown Primary
TMPRSS2 (x26) – Ets-domain retained – Prostate Cancer
TMPRSS2 (x8) – Ets-domain lost – Prostate Cancer
EWSR1 (x7) – Ets-domain retained – Bone Cancer, Soft Tissue Sarcoma
TMPRSS2 (x6) – Ets-domain lost – Prostate Cancer
EWSR1 (x6) – Ets-domain retained – Bone Cancer
TMPRSS2 – Ets-domain lost – Prostate Cancer
TMPRSS2 – Ets-domain retained – Prostate Cancer
TMPRSS2 – Ets-domain retained – Prostate Cancer
TMPRSS2 – Ets-domain lost – Prostate Cancer
TNK1 – Ets-domain retained – Breast Cancer
UBASH3A – Ets-domain retained – Breast Cancer
EWSR1 – Ets-domain retained – Bone Cancer
NETO1 – Ets-domain retained – Non-Small Cell Lung Cancer
EWSR1 – Ets-domain retained – Bone Cancer
EWSR1 – Ets-domain retained – Bone Cancer
EWSR1 – Ets-domain retained – Bone Cancer
RBBP8 – Ets-domain retained – Head and Neck Cancer
GABPA – Ets-domain truncated – Soft Tissue Sarcoma
FLI1 – Ets-domain truncated – Esophagogastric Cancer

Partner-block colors are decorative; row labels identify partners.

Full per-gene detail: *gene_reports/erg/report.md*

Discussion

- Cross-gene Benjamini-Hochberg FDR correction was applied jointly across the 359 genes that produced at least one computable p-value in this scan. This reduces false-positive findings relative to testing each gene in isolation, but is a conservative correction: a real per-gene effect can fail to reach the $q < 0.05$ threshold once corrected across that full p-value-bearing gene set.
- 10 gene(s) used hand-curated gene configurations, while 514 gene(s) were auto-configured from Genome Nexus canonical-transcript/Pfam-domain data using a kinase/catalytic-keyword heuristic to select the tracked domain; auto-configured domains have not been manually verified the way hand-curated ones have.
- Data were retrieved live from the public msk_impact_50k_2026 cBioPortal study as of 2026-09-15T16:38:12.426581+00:00. Some cBioPortal cohorts (e.g. actively accruing clinical-sequencing panels such as MSK-IMPACT) are updated periodically, so exact counts could shift if this scan is re-run later against such a cohort.
- All findings in this manuscript are computational, hypothesis-generating candidate evidence from bioinformatic analysis of public cohort data, not validated clinical calls; a gene's presence here is not a therapeutic or diagnostic recommendation.

Appendix: per-gene report index

Gene	Tier	Config source	N events	Best q	Full report
NTRK3	FDR-significant; Curated gene config	curated	58	9.002e-09	gene_reports/ntrk3/report.md
ROS1	FDR-significant; Curated gene config	curated	122	0.000244	gene_reports/ros1/report.md
ETV6	FDR-significant; Curated gene config	curated	90	0.002572	gene_reports/etv6/report.md
FLI1	FDR-significant; Curated gene config	curated	118	0.009047	gene_reports/fli1/report.md
RET	Honorable mention; Curated gene config	curated	194	0.1066	gene_reports/ret/report.md
FGFR2	Honorable mention; Curated gene config	curated	136	0.1066	gene_reports/fgfr2/report.md
ALK	Honorable mention; Curated gene config	curated	272	0.1675	gene_reports/alk/report.md
EGFR	Honorable mention	auto	55	0.1917	gene_reports/egfr/report.md
BRAF	Honorable mention; Curated gene config	curated	179	0.1675	gene_reports/braf/report.md
FGFR3	Honorable mention	auto	152	0.1675	gene_reports/fgfr3/report.md
CDKN2B	Honorable mention	auto	12	0.291	gene_reports/cdkn2b/report.md
IKBKE	Honorable mention	auto	12	0.2663	gene_reports/ikbke/report.md
TMPRSS2	Honorable mention	auto	867	0.1675	gene_reports/tmpRSS2/report.md
NTRK1	Honorable mention; Curated gene config	curated	78	0.1675	gene_reports/ntrk1/report.md
INPPL1	Honorable mention	auto	14	0.3846	gene_reports/inppl1/report.md
CDK12	Honorable mention	auto	43	0.1675	gene_reports/cdk12/report.md
NAB2	Honorable mention	auto	72	0.1675	gene_reports/nab2/report.md
KDM5A	Honorable mention	auto	10	0.7728	gene_reports/kdm5a/report.md
FH	Honorable mention	auto	16	0.5044	gene_reports/fh/report.md
ERG	Curated gene config	curated	788	0.1675	gene_reports/erg/report.md